

Chapter 27

Agradecimientos

Este libro es un resultado de trabajo en común y el esfuerzo de muchas personas. Aprecio a todas las personas que han contribuido a este libro, ya sea con fotos, revisiones, o ayudando a entrar datos en COMPADRE y especialmente a los colaboradores. La información resumida en este libro no habría sido posibles sin las cientos y miles de horas que los estudiantes y colegas han dedicado a sus investigaciones y publicaciones. Con los resultados presentado en los artículos científicos y tesis hemos podido comenzar a entender y apreciar la diversidad de patrones de la historia de vida de las especies de orquídeas y comenzar a evaluar como el estudio de la historia de vida puede ayudarnos a evaluar la el riesgo de la supervivencia de la poblaciones y especies por causa natural o efecto antropogénico. El futuro también se ve muy interesante en que podemos comenzar a evaluar patrones que pudiese describir como la historia de vida de la orquídeas se parece y difiere de otros grupos taxonómicos y a lo mejor describir un otro angulo de por que la familia de las Orquídeas es tan abundante.

27.1 Fotos

Aprecio las siguientes personas por haber compartido fotos

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Fotos=tribble(~"Origen de las Fotos", ~Especie, ~ Dirección,
              "Diana Molina Ozuma", "*Erycina crista-galli*", "Departamento de Ecología
              "James D. Ackerman", "*Dactylorhiza maculata*", *D. sambucina*", "Departmen
              "Eduardo A. Pérez García", "*Laelia speciosa*", "Universidad Nacional Autó
              "Hans Jacquemyn", "*Orchis purpurea*", "Leuven Plant Institute, KU Leuven,
              "Giuseppe Pellegrino", "*Serapias cordigera*", "Ecology and Earth Sciences
              "Mitch McClaran", "*Spiranthes delistecens*", "Professor of Range Manageme
              "Hong Liu", "*Trichocentrum undulatum*", "Department of Earth and Environment, Florida I
              "Edwin Guevara", "*Lepanthes caritensis* \n *Lepanthes eltoroensis* \n *Lepanthes woodbu
              "Mark Blackman", "*Vanilla planifolia*", "Avenida del IMAN111-601, Insurgentes-Cuiculco,
```

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"Leonel López-Toledo", "*Laelia speciosa*", "Instituto de Investigaciones sobre los Recursos Naturales",
"Alonso Argüero", "*Psychilis karwinskii*", "Departamento de Ecología y Recursos Naturales",
"Claudia C. Guitiérrez", "*Cypripedium irapeanum*", "27 A Oriente 1812-1, Fraccionamiento San Antonio",
)
```

```
Fotos_out=Fotos %>%
  flextable() %>%
  theme_zebra() %>%
  theme_box() |>

  set_table_properties(layout = "autofit")

Fotos_out
```

Origen de las Fotos	Especie	Dirección
Diana Molina Ozuma	*Erycina crista-galli*	Departamento de Ecología y Recursos Naturales
James D. Ackerman	*Dactylorhiza maculata*, *D. sambucina*	Departamento de Ecología y Recursos Naturales
Eduardo A. Pérez García	*Laelia speciosa*	Universidad de Leuven
Hans Jacquemyn	*Orchis purpurea*	Leuven
Giuseppe Pellegrino	*Serapias cordigera*	Ecology
Mitch McClaran	*Spiranthes delistecens*	Profesor
Hong Liu	*Trichocentrum undulatum*	Departamento de Ecología y Recursos Naturales
Edwin Guevara	*Lepanthes caritensis* *Lepanthes eltoroensis* *Lepanthes woodburyana*	Maryland
Mark Blackman	*Vanilla planifolia*	Avenida
Leonel López-Toledo	*Laelia speciosa*	Instituto de Investigaciones sobre los Recursos Naturales
Alonso Argüero	*Psychilis karwinskii*	Departamento de Ecología y Recursos Naturales
Claudia C. Guitiérrez	*Cypripedium irapeanum*	27 A Oriente

27.2 Revisores

Las siguientes personas ayudaron a revisar algunos de los capítulos. Todos los errores son míos y no de ellos.

```
Revisores=tribble(~"Nombre", ~"Institución",
  "Joel Tupac Otero", "Departamento de Ciencias Biológicas, Universidad Naci
  "Maria Eglée Pérez", "Departamento de Matemática, Universidad de Puerto Ri
  "Hibraim Adán Pérez Mendoza", "Facultad de Estudios Superiores-Iztacala UN
  )
```

```
Revisores_out=Revisores %>%
  flextable() %>%
  theme_zebra() %>%
  theme_box() |>

  set_table_properties(layout = "autofit")

Revisores_out
```

Nombre	Institución
Joel Tupac Otero	Departamento de Ciencias Biológicas, Universidad Nacional de Colombia, Sede
Maria Eglée Pérez	Departamento de Matemática, Universidad de Puerto Rico, San Juan, Puerto R
Hibraim Adán Pérez Mendoza	Facultad de Estudios Superiores-Iztacala UNAM, México

27.3 Entrada de datos en COMPADRE

Aprecio las siguientes personas por haber entrado datos de PPM de orquídeas en COMPADRE

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Benevolos=tribble(~"Nombre", ~"Institución",
  "Javielise Donato", "Universidad de Puerto Rico-Humacao, Puerto Rico",
  "Diana Clausia Molina Ozuna", "Conservación de la Biodiversidad, El Colegi
  "Mayliz M. Ortiz Ortiz", "Universidad de Puerto Rico-Humacao, Puerto Rico"
  "Gabriela Torres Torres", "Universidad del Valle, Colombia")
```

```
Benevolos_out=Benevolos %>%
  flextable() %>%
  theme_zebra() %>%
  theme_box() |>

  set_table_properties(layout = "autofit")
```

Benevolos_out

Nombre	Institución
Javielise Donato	Universidad de Puerto Rico-Humacao, Puerto Rico
Diana Clausia Molina Ozuna	Conservación de la Biodiversidad, El Colegio de la Frontera Sur, ECOSUR, México
Nayliz M. Ortiz Ortiz	Universidad de Puerto Rico-Humacao, Puerto Rico
Gabriela Torres Torres	Universidad del Valle, Colombia

- Acácio V, Holmgren M, Jansen PA, Schrotter O. 2007. Multiple recruitment limitation causes arrested succession in Mediterranean cork oak systems. *Ecosystems* 10:1220-1230.
- Ackerman JD. 1989. Limitations to sexual reproduction in *Encyclia krugii* (Orchidaceae). *Systematic Botany*:101-109.
- Ackerman JD, Phillips RD, Tremblay RL, Karremans A, Reiter N, Peter CI, Bogarín D, Pérez-Escobar OA, Liu H. 2023. Beyond the various contrivances by which orchids are pollinated: global patterns in orchid pollination biology. *Botanical Journal of the Linnean Society* 202:295-324.
- Ackerman JD, Sabat A, Zimmerman J. 1996. Seedling establishment in an epiphytic orchid: an experimental study of seed limitation. *Oecologia* 106:192-198.
- Ackerman JD, Tremblay R, Arias T, Zotz G, Sharma J, Salazar GA, Kaur J. 2025. Persistent Habitat Instability and Patchiness, Sexual Attraction, Founder Events, Drift and Selection: A Recipe for Rapid Diversification of Orchids. *Plants* 14:1193.
- Ackerman J, Tremblay R, Pérez M, Madden H, Bechtold M, Boeken M. 2020. Small populations on small islands: What chance does an orchid have? *International Journal of Plant Sciences* 181:667-685.
- Adler PB, Salguero-Gómez R, Compagnoni A, Hsu JS, Ray-Mukherjee J, Mbeau-Ache C, Franco M. 2014. Functional traits explain variation in plant life history strategies. *Proceedings of the National Academy of Sciences* 111:740-745.
- Aide TM, Clark ML, Grau HR, López-Carr D, Levy MA, Redo D, Bonilla-Moheno M, Riner G, Andrade-Núñez MJ, Muñiz M. 2013. Deforestation and Reforestation of Latin America and the Caribbean (2001–2010). *Biotropica* 45:262-271.
- Alexander C, Alexander I. 1984. Seasonal changes in populations of the orchid *Goodyera repens* Br. and in its mycorrhizal development. In: *Transactions of the Botanical Society of Edinburgh*. Taylor & Francis, 219-227.
- Alves JM, Carneiro M, Day JP, Welch JJ, Duckworth JA, Cox TE, Letnic M, Strive T, Ferrand N, Jiggins FM. 2022. A single introduction of wild rabbits triggered the biological invasion of Australia. *Proceedings of the National Academy of Sciences* 119:e2122734119.
- Anderson DR, Burnham KP. 2002. Avoiding pitfalls when using information-theoretic methods. *The Journal of Wildlife Management*:912-918.

- Anghelescu NE, Vafae Y, Ahmadzadeh K, Chen J-T. 2023. Asymbiotic seed germination in terrestrial orchids: problems, progress, and prospects. *Advances in Orchid Biology, Biotechnology and Omics*:221-260.
- Antonini Y, Dirzo R, Cassia Quitete-Portela R de. 2020. Are protected populations of two globular cactus species facing a demographic explosion or just a «bonanza» year? *Journal of Arid Environments* 179:104192.
- Aragón S, Ackerman JD. 2004. Does flower color variation matter in deception pollinated *Psychilis monensis* (Orchidaceae)? *Oecologia* 138:405-413.
- Archer CR, Paniw M, Vega-Trejo R, Sepil I. 2022. A sex skew in life-history research: the problem of missing males. *Proceedings of the Royal Society B* 289:20221117.
- Arditti J. 1967. Factors affecting the germination of orchid seeds. *The Botanical Review* 33:1-97.
- Arditti J, Ghani A. 2000. Numerical and physical properties of orchid seeds and their biological implications (vol 145, pg 367, 2000). *New Phytologist* 146:569-569.
- Armbruster P, Lande R. 1993. A population viability analysis for African elephant (*Loxodonta africana*): how big should reserves be? *Conservation Biology* 7:602-610.
- Arroyo-Cosultchi G, Golubov J, Solórzano JV, Mandujano MC. 2022. Prescriptions for the Control of a Clonal Invasive Species Using Demographic Models. *Plants* 11:689.
- Aucencia E-L, Anne D. 2024. Acclimatization with endophytic fungi and reintroduction of *Guarianthe skinneri* (Bateman) Dressler & WE Higgins, a threatened, native orchid of cultural value in southern Mexico. *Journal for Nature Conservation* 78:126573.
- Bacaër N. 2011. *A Short History of Mathematical Population Dynamics*. Springer.
- Baden HM, Roach D, Schweingruber FH, Reitzel K, Lundgreen K, Dahlgren JP. 2021. The effects of age on the demography of a perennial plant depend on interactions with size and environment. *Journal of Ecology* 109:1068-1077.
- Baker JD, Tate GR. 1867. *A new flora of Northumberland and Durham*. Tyneside Naturalists' Field Club, London.
- Balducci MG, Calevo J, Duffy KJ. 2024. Narrow mycorrhizal specialization and its effect on early development in the Mediterranean orchid *Orchis italica*. *Botanical Journal of the Linnean Society*:boae018.
- Barreda-Castillo JM, Pansarin ER, Monribot-Villanueva JL, Guerrero-Analco JA, Menchaca-García RA. 2024. Allogamy, metaxenia and hybrids in orchids, what do we know about it? *Botanical Sciences* 102:646-670.
- Bateman RM. 2020. Implications of next-generation sequencing for the systematics and evolution of the terrestrial orchid genus *Epipactis*, with particular reference to the British Isles. *Kew Bulletin* 75:4.
- Batty A, Brundrett M, Dixon K, Sivasithamparam K. 2006. In situ symbiotic seed germination and propagation of terrestrial orchid seedlings for establishment at field sites. *Australian Journal of Botany* 54:375-381.
- Batty A, Dixon K, Brundrett M, Sivasithamparam K. 2001. Constraints to symbiotic germination of terrestrial orchid seed in a mediterranean bushland. *New Phytologist* 152:511-520.
- Baudisch A, Salguero-Gómez R, Jones OR, Wrycza T, Mbeau-Ache C, Franco M, Colchero F. 2013. The pace and shape of senescence in angiosperms. *Journal of Ecology* 101:596-606.
- Beissinger SR, Westphal MI. 1998. On the use of demographic models of population viability

- in endangered species management. *The Journal of Wildlife Management*:821-841.
- Bell T, Bowles M, McEachern A. 2003. Projecting the success of plant population restoration with viability analysis. In: *Population viability in plants: Conservation, management, and modeling of rare plants*. Springer, 313-348.
- Bellis J, Osazuwa-Peters O, Maschinski J, Keir MJ, Parsons EW, Kaye TN, Kunz M, Possley J, Menges E, Smith SA, others. 2024. Identifying predictors of translocation success in rare plant species. *Conservation Biology* 38:e14190.
- Benzing D. 1981. Bark surfaces and the origin and maintenance of diversity among angiosperm epiphytes: a hypothesis. *Selbyana* 5:248-255.
- Bernard C, Santos GS, Deere JA, Rodriguez-Caro R, Capdevila P, Kusch E, Gascoigne SJ, Jackson J, Salguero-Gómez R. 2023. MOSAIC-A unified trait database to complement structured population models. *Scientific Data* 10:335.
- Berry EJ, Cleavitt NL. 2021. Population dynamics and comparative demographics in sympatric populations of the round-leaved orchids *Platanthera macrophylla* and *P. orbiculata*. *Population Ecology* 63:274-289.
- Bialic-Murphy L, Gaoue OG, Knight T. 2018. Using Transfer Function Analysis to develop biologically and economically efficient restoration strategies. *Scientific Reports* 8:2094.
- Bierzychudek P. 1999. Looking backwards: assessing the projections of a transition matrix model. *Ecological Applications* 9:1278-1287.
- Birch L. 1953. Experimental background to the study of the distribution and abundance of insects: I. The influence of temperature, moisture and food on the innate capacity for increase of three grain beetles. *Ecology* 34:698-711.
- Bock MJ, Jarvis GC, Corey EL, Stone EE, Gribble KE. 2019. Maternal age alters offspring lifespan, fitness, and lifespan extension under caloric restriction. *Scientific Reports* 9:3138.
- Borges JL. 1999. *Collected fictions*. Penguin.
- Borrero H, Oviedo-Prieto R, Alvarez JC, Ticktin T, Cisneros M, Liu H. 2023. Populations of a tropical epiphytic orchid are destabilized in its peripheral range by hurricane and an exotic herbivore. *Ecosphere* 14:e4355.
- Boukal DS, Berec L. 2002. Single-species models of the Allee effect: extinction boundaries, sex ratios and mate encounters. *Journal of Theoretical Biology* 218:375-394.
- Bradshaw ME, Doody J. 1978. Plant population studies and their relevance to nature conservation. *Biological Conservation* 14:223-242.
- Bruijning M, Jongejans E, Turcotte MM. 2019. Demographic responses underlying eco-evolutionary dynamics as revealed with inverse modelling. *Journal of Animal Ecology* 88:768-779.
- Buckley YM, Brieze DT, Rees M. 2003. Demography and management of the invasive plant species *Hypericum perforatum*. II. Construction and use of an individual-based model to predict population dynamics and the effects of management strategies. *Journal of Applied Ecology* 40:494-507.
- Buckley YM, Ramula S, Blomberg SP, Burns JH, Crone EE, Ehrlén J, Knight TM, Pichancourt J-B, Quested H, Wardle GM. 2010. Causes and consequences of variation in plant population growth rate: a synthesis of matrix population models in a phylogenetic context. *Ecology Letters* 13:1182-1197.
- Burd M. 1995. Pollinator behavioural responses to reward size in *Lobelia deckenii*: no escape

- from pollen limitation of seed set. *Journal of Ecology*:865-872.
- Bustamante J. 1996. Population viability analysis of captive and released bearded vulture populations. *Conservation Biology* 10:822-831.
- Caballero-Villalobos L, Silva-Arias GA, Buzatto CR, Nervo MH, Singer RB. 2017. Generalized food-deceptive pollination in four *Cattleya* (Orchidaceae: Laeliinae) species from Southern Brazil. *Flora* 234:195-206.
- Callaway RM, Reinhart KO, Moore GW, Moore DJ, Pennings SC. 2002. Epiphyte host preferences and host traits: mechanisms for species-specific interactions. *Oecologia* 132:221-230.
- Calvo RN. 1993. Evolutionary demography of orchids: intensity and frequency of pollination and the cost of fruiting. *Ecology* 74:1033-1042. DOI: [10.2307/1940473](https://doi.org/10.2307/1940473).
- Calvo RN, Horvitz CC. 1990. Pollinator limitation, cost of reproduction, and fitness in plants: a transition-matrix demographic approach. *The American Naturalist* 136:499-516.
- Capdevila P, Stott I, Beger M, Salguero-Gómez R. 2020. Towards a comparative framework of demographic resilience. *Trends in Ecology & Evolution* 35:776-786.
- Caplat P, Coutts S, Buckley YM. 2012. Modeling population dynamics, landscape structure, and management decisions for controlling the spread of invasive plants. *Annals of the New York Academy of Sciences* 1249:72-83.
- Carey P. 1998. Modelling the spread of *Himantoglossum hircinum* (L.) Spreng. at a site in the south of England. *Botanical Journal of the Linnean Society* 126:159-172.
- Case MA, Bradford ZR. 2009. Enhancing the trap of lady's slippers: a new technique for discovering pollinators yields new data from *Cypripedium parviflorum* (Orchidaceae). *Botanical Journal of the Linnean Society* 160:1-10.
- Caswell H. 1989. Analysis of life table response experiments I. Decomposition of effects on population growth rate. *Ecological Modelling* 46:221-237.
- Caswell H. 2000. *Matrix population models*. Sinauer Sunderland, MA.
- Caswell H. 2001. *Matrix Population Models: Construction, analysis, and interpretation*. Sunderland, Massachusetts, USA: Sinauer Associates.
- Caswell H. 2009. Stage, age and individual stochasticity in demography. *Oikos* 118:1763-1782.
- Caswell H. 2010. Life table response experiment analysis of the stochastic growth rate. *Journal of Ecology* 98:324-333.
- Caswell H. 2011. Beyond R 0: Demographic models for variability of lifetime reproductive output. *PloS One* 6:e20809.
- Caswell H. 2012. Matrix models and sensitivity analysis of populations classified by age and stage: a vec-permutation matrix approach. *Theoretical Ecology* 5:403-417.
- Caswell H. 2019a. The formal demography of kinship: A matrix formulation. *Demographic Research* 41:679-712. DOI: [10.4054/DemRes.2019.41.24](https://doi.org/10.4054/DemRes.2019.41.24).
- Caswell H. 2019b. *Sensitivity analysis: matrix methods in demography and ecology*. Springer Nature.
- Caswell H. 2020. The formal demography of kinship ii. *Demographic Research* 42:1097-1144.
- Caswell H, Brault S, Read AJ, Smith TD. 1998. Harbor porpoise and fisheries: an uncertainty analysis of incidental mortality. *Ecological Applications* 8:1226-1238.
- Caswell H, Daalen S van. 2021. Healthy longevity from incidence-based models.

- Demographic Research* 45:397-452.
- Caswell H, Salguero-Gómez R. 2013. Age, stage and senescence in plants. *Journal of Ecology* 101:585-595.
- Caswell H, Vries C de, Hartemink N, Roth G, Daalen SF van. 2018. Age $\times$ stage-classified demographic analysis: a comprehensive approach. *Ecological Monographs* 88:560-584.
- Catling PM. 1986. Epiphytic orchids in a Belizean grapefruit orchard: distribution, colonization, and association. *Lindleyana* 1:194-202.
- Catling PM. 1990. Auto-pollination in the Orchidaceae. In: Arditti J ed. *Orchid Biology: Reviews and Perspectives V*. Timber Press, Portland, Oregon, 123-158.
- Chase MW. 1987. Obligate twig epiphytism in the Oncidiinae and other neotropical orchids. *Selbyana*:24-30.
- Che-Castaldo J, Havercamp K, Watanuki K, Matsuzawa T, Hirata S, Ross SR. 2021. Comparative survival analyses among captive chimpanzees (*Pan troglodytes*) in America and Japan. *PeerJ* 9:e11913.
- Che-Castaldo J, Jones OR, Kendall BE, Burns JH, Childs DZ, Ezard TH, Hernandez-Yanez H, Hodgson DJ, Jongejans E, Knight T, others. 2020. Comments to «Persistent problems in the construction of matrix population models».
- Childs DZ, Sheldon BC, Rees M. 2016. The evolution of labile traits in sex-and age-structured populations. *Journal of Animal Ecology* 85:329-342.
- Choquet R, Rouan L, Pradel R. 2009. Program E-SURGE: a software application for fitting multievent models. *Modeling demographic processes in marked populations*:845-865.
- Chung MY, Nason JD, Chung MG. 2004. Spatial genetic structure in populations of the terrestrial orchid *Cephalanthera longibracteata* (Orchidaceae). *American Journal of Botany* 91:52-57.
- Clubb R, Mason G. 2003. Captivity effects on wide-ranging carnivores. *Nature* 425:473-474.
- Clubb R, Rowcliffe M, Lee P, Mar KU, Moss C, Mason GJ. 2009. Fecundity and population viability in female zoo elephants: problems and possible solutions. *Animal Welfare* 18:237-247.
- Coates F, Lunt ID, Tremblay R. 2006. Effects of disturbance on population dynamics of the threatened orchid *Prasophyllum correctum* DL Jones and implications for grassland management in south-eastern Australia. *Biological Conservation* 129:59-69.
- Cochran ME, Ellner S. 1992. Simple Methods for Calculating Age-Based Life History Parameters for Stage-Structured Populations: Ecological Archives M062-002. *Ecological Monographs* 62:345-364.
- Compagnoni A, Steigman K, Miller TE. 2017. Can't live with them, can't live without them? Balancing mating and competition in two-sex populations. *Proceedings of the Royal Society B: Biological Sciences* 284:20171999.
- Conde DA, Staerk J, Colchero F, Silva R da, Schöley J, Baden HM, Jouvét L, Fa JE, Syed H, Jongejans E, others. 2019. Data gaps and opportunities for comparative and conservation biology. *Proceedings of the National Academy of Sciences* 116:9658-9664.
- Connolly J, Wayne P. 1996. Asymmetric competition between plant species. *Oecologia* 108:311-320.
- Cooch EG, Gauthier G, Rockwell RF. 2003. Apparent differences in stochastic growth rates based on timing of census: a cautionary note. *Ecological Modelling* 159:133-143.
- Cook RE. 1983. Clonal plant populations: A knowledge of clonal structure can affect the

- interpretation of data in a broad range of ecological and evolutionary studies. *American Scientist* 71:244-253.
- Coste CF, Pavard S. 2020. Analysis of a multitrait population projection matrix reveals the evolutionary and demographic effects of a life history trade-off. *Ecological Modelling* 418:108915.
- Coulson T, Catchpole EA, Albon SD, Morgan BJ, Pemberton J, Clutton-Brock TH, Crawley M, Grenfell BT. 2001. Age, sex, density, winter weather, and population crashes in Soay sheep. *Science* 292:1528-1531.
- Crain B, Tremblay R, Ferguson J. 2019. Sheltered from the storm? Population viability analysis of a rare endemic under periodic catastrophe regimes. *Population Ecology* 61:74-92. DOI: [10.1002/1438-390X.1002](https://doi.org/10.1002/1438-390X.1002).
- Crawley MJ. 1990. The population dynamics of plants. *Philosophical Transactions of the Royal Society of London. Series B: Biological Sciences* 330:125-140.
- Cribb PJ. 1999. II Morphology. *Genera Orchidacearum: Volume 1: Apostasioideae and Cypripedioideae* 11:12.
- Crone EE, Menges ES, Ellis MM, Bell T, Bierzychudek P, Ehrlén J, Kaye TN, Knight TM, Lesica P, Morris WF, others. 2011. How do plant ecologists use matrix population models? *Ecology Letters* 14:1-8.
- Crouse DT, Crowder LB, Caswell H. 1987. A stage-based population model for loggerhead sea turtles and implications for conservation. *Ecology* 68:1412-1423.
- Crowder LB, Crouse DT, Heppell SS, Martin TH. 1994. Predicting the impact of turtle excluder devices on loggerhead sea turtle populations. *Ecological Applications* 4:437-445.
- Cushing JM. 2003. *Chaos in ecology: experimental nonlinear dynamics*. Elsevier.
- D'Agostino M, Cao Pinna L, Carboni M, Assini S, Bacchetta G, Bartolucci F, Brancaloni L, Buldrini F, Carta A, Cerabolini B, others. 2024. Best practices, errors, and perspectives of half a century of plant translocation in Italy. *Conservation Biology* 38:e14233.
- Daalen SF van, Caswell H. 2017. Lifetime reproductive output: individual stochasticity, variance, and sensitivity analysis. *Theoretical Ecology* 10:355-374.
- Daalen SF van, Hernández CM, Caswell H, Neubert MG, Gribble KE. 2022. The contributions of maternal age heterogeneity to variance in lifetime reproductive output. *The American Naturalist* 199:603-616.
- Dafni A. 1983. Pollination of *Orchis caspia*—a nectarless plant which deceives the pollinators of nectariferous species from other plant families. *The Journal of Ecology*:467-474.
- Darwin C. 1877. *The various contrivances by which orchids are fertilised by insects*. John Murray.
- Davis KJ. 2022. Managed culls mean extinction for a marine mammal population when combined with extreme climate impacts. *Ecological Modelling* 473:110122.
- Davison R, Jacquemyn H, Adriaens D, Honnay O, De Kroon H, Tuljapurkar S. 2010. Demographic effects of extreme weather events on a short-lived calcareous grassland species: stochastic life table response experiments. *Journal of Ecology* 98:255-267.
- De Kroon H, Van Groenendael J, Ehrlén J. 2000. Elasticities: a review of methods and model limitations. *Ecology* 81:607-618.
- De Magalhaes J, Costa. 2009. A database of vertebrate longevity records and their relation to other life-history traits. *Journal of evolutionary biology* 22:1770-1774.
- Demetrius L. 1992. Growth rate, population entropy, y evolutionary dynamics. *Theoretical*

- Population Biology* 41:208-236.
- Doak DF, Diane Thomson, Jules ES. 2002. *Population Viability Analysis for Plants: Understanding the Demographic Consequences of Seed Banks for Population Health*. University of Chicago Press.
- Doak DF, Morris WF, Pfister C, Kendall BE, Bruna EM. 2005. Correctly estimating how environmental stochasticity influences fitness and population growth. *The American Naturalist* 166:E14-E21.
- Dressler RL. 1993. *Phylogeny and classification of the orchid family*. Cambridge University Press.
- Drugs UNO on, Crime. 2024. *World Wildlife Crime Report 2024: Trafficking in Protected Species*. UN.
- Earl JE. 2019. Evaluating the assumptions of population projection models used for conservation. *Biological Conservation* 237:145-154.
- Ebert thomas A. 1999. *Plant and Animal Populations. Methods in Demography*. Academic Press.
- Eckert CG, Herlihy CR. 2004. Using a cost-benefit approach to understand the evolution of self-fertilization in plants: the perplexing case of *Aquilegia canadensis* (Ranunculaceae). *Plant Species Biology* 19:159-173.
- Ehrlén J, Eriksson O. 2000. Dispersal limitation and patch occupancy in forest herbs. *Ecology* 81:1667-1674.
- Ehrlén J, Morris WF, Euler T von, Dahlgren JP. 2016. Advancing environmentally explicit structured population models of plants. *Journal of Ecology* 104:292-305.
- Ehrlén J, Van Groenendael J. 1998. Direct perturbation analysis for better conservation. *Conservation Biology* 12:470-474.
- Ellner SP, Childs DZ, Rees M, others. 2016. Data-driven modelling of structured populations. *A practical guide to the Integral Projection Model*. Cham: Springer.
- Emery SM, Gross KL. 2005. Effects of timing of prescribed fire on the demography of an invasive plant, spotted knapweed *Centaurea maculosa*. *Journal of Applied Ecology* 42:60-69.
- Emeterio-Lara A, García-Franco JG, Hernández-Apolinar M, Toledo-Hernández VH, Valencia-Díaz S, Flores-Palacios A. 2021. Does extraction of orchids affect their population structure? Evidence from populations of *Laelia autumnalis* (Orchidaceae). *Forest Ecology and Management* 480:118667.
- Emeterio-Lara A, Palma-Linares V, Vázquez-García LM, Mejía-Carranza J. 2016. Usos y comercialización de orquídeas silvestres en la región sur del Estado de México. *Polibotánica*:197-214.
- Evans A, Jacquemyn H. 2020. Impact of mating system on range size and niche breadth in *Epipactis* (Orchidaceae). *Annals of Botany* 126:1203-1214.
- Ezard TH, Bullock JM, Dalglish HJ, Millon A, Pelletier F, Ozgul A, Koons DN. 2010. Matrix models for a changeable world: the importance of transient dynamics in population management. *Journal of Applied Ecology* 47:515-523.
- Falcón W, Ackerman JD, Tremblay R. 2017. Quantifying how acquired interactions with native and invasive insects influence population growth rates of a non-indigenous plant. *Biological Invasions* 19:895-911. DOI: [10.1007/s10530-016-1318-8](https://doi.org/10.1007/s10530-016-1318-8).
- Fenner M. 2017. Ecology of seed banks. In: *Seed development and germination*. Routledge,

507-528.

- Ferreira MS, Kajin M, Cerqueira R, Vieira MV. 2016. Marsupial population dynamics in a tropical rainforest: intraspecific competition and nonlinear effect of rainfall. *Journal of Mammalogy* 97:121-127.
- Flather CH, Hayward GD, Beissinger SR, Stephens PA. 2011. Minimum viable populations: is there a 'magic number' for conservation practitioners? *Trends in Ecology & Evolution* 26:307-316.
- Forsman E, Sovern S, Seaman DE, Maurice K, Taylor M, Zisa J. 1996. Demography of the northern spotted owl on the Olympic Peninsula and east slope of the Cascade Range, Washington. *Studies in Avian Biology* 17:21-30.
- Franco M, Silvertown J. 1996. Life history variation in plants: an exploration of the fast-slow continuum hypothesis. *Philosophical Transactions of the Royal Society of London. Series B: Biological Sciences* 351:1341-1348.
- Franco M, Silvertown J. 2004. A comparative demography of plants based upon elasticities of vital rates. *Ecology* 85:531-538.
- Freckleton R, Watkinson A. 2001. Asymmetric competition between plant species. *Functional Ecology* 15:615-623.
- Fricke EC, Tewksbury JJ, Rogers HS. 2019. Linking intra-specific trait variation and plant function: seed size mediates performance tradeoffs within species. *Oikos* 128:1716-1725.
- Fujiwara M, Caswell H. 2001. Demography of the endangered North Atlantic right whale. *Nature* 414:537-541.
- Gabriel A, John R, others. 2017. On the rarity of dioecy in flowering plants. *Molecular Ecology* 26.
- Gaoue OG, Ticktin T. 2010. Effects of harvest of nontimber forest products and ecological differences between sites on the demography of African mahogany. *Conservation Biology* 24:605-614.
- García MB, Ehrlén J. 2002. Reproductive effort and herbivory timing in a perennial herb: fitness components at the individual and population levels. *American Journal of Botany* 89:1295-1302.
- García MB, Goni D, Guzmán D. 2010. Living at the edge: Local versus positional factors in the long-term population dynamics of an endangered orchid. *Conservation Biology* 24:1219-1229.
- García-Soriano R. 2003. Demografía, manejo y conservación de *Artorima erubescens* en Oaxaca, orquídea endémica del sur de México. mathesis Thesis. Universidad Autonoma de Yucatan.
- Gascoigne SJ, Rolph S, Sankey D, Nidadavolu N, Stell Pičman AS, Hernández CM, Philpott ME, Salam A, Bernard C, Fenollosa E, others. 2023. A standard protocol to report discrete stage-structured demographic information. *Methods in Ecology and Evolution* 14:2065-2083.
- Gatsuk L, Smirnova O, Vorontzova L, Zaugolnova L, Zhukova L. 1980. Age states of plants of various growth forms: a review. *The Journal of Ecology*:675-696.
- Gerlach Jr JD, Rice KJ. 2003. Testing life history correlates of invasiveness using congeneric plant species. *Ecological Applications* 13:167-179.
- Gerstner K, Moreno-Mateos D, Gurevitch J, Beckmann M, Kambach S, Jones HP, Seppelt R. 2017. Will your paper be used in a meta-analysis? Make the reach of your research

- broader and longer lasting. *Methods in Ecology and Evolution* 8:777-784.
- Ghimire SK, Gimenez O, Pradel R, McKey D, Aumeeruddy-Thomas Y. 2008. Demographic variation and population viability in a threatened Himalayan medicinal and aromatic herb *Nardostachys grandiflora*: matrix modelling of harvesting effects in two contrasting habitats. *Journal of Applied Ecology* 45:41-51.
- Golubov J, Mandujano MC. 2009. Por qué los polinizadores visitan a las flores. *Revista bimestral Casa del Tiempo* 21:39-41.
- Gontijo LM, Carvalho RM. 2020. Using life stage-structured matrix models to determine natural enemy: pest release ratios for augmentative biological control. *Journal of Applied Entomology* 144:364-372.
- González-Orellana N, Mendoza AS, Tremblay R, Ackerman J. 2024. Host suitability for germination differs from that of later stages of development in a rare epiphytic orchid. *Lankesteriana: International Journal on Orchidology*:93-114.
- Goode LK, Allen MF. 2008. The impacts of Hurricane Wilma on the epiphytes of El Edén Ecological Reserve, Quintana Roo, Mexico. *The Journal of the Torrey Botanical Society* 135:377-387.
- Gotelli NJ, Ellison AM. 2006. Forecasting extinction risk with nonstationary matrix models. *Ecological Applications* 16:51-61.
- Gould WA, Wadsworth FH, Quiñones M, Fain SJ, Álvarez-Berrios NL. 2017. Land use, conservation, forestry, and agriculture in Puerto Rico. *Forests* 8:242.
- Gregg KB. 1989. Reproductive biology of the orchid *Cleistes divaricata* (L.) Ames var. *bifaria* Fernald growing in a West Virginia meadow. *Castanea*:57-78.
- Gregg KB. 1991. Variation in Behaviour of four populations of the orchid *Cleistes divaricata*, an assessment using transition matrix models. In: Wells TCE, Willems JH eds. *Population Ecology of Terrestrial Orchids*. SPB Academic Publishing bv, The Hague, 139-159.
- Gregg KB, Kéry M. 2006. Comparison of size vs. life-state classification in demographic models for the terrestrial orchid *Cleistes bifaria*. *Biological Conservation* 129:50-58.
- Gremer JR, Sala A. 2013. It is risky out there: the costs of emergence and the benefits of prolonged dormancy. *Oecologia* 172:937-947.
- Grindeland JM, Sletvold N, Ims RA. 2005. Effects of floral display size and plant density on pollinator visitation rate in a natural population of *Digitalis purpurea*. *Functional Ecology*:383-390.
- Gurevitch J, Koricheva J, Nakagawa S, Stewart G. 2018. Meta-analysis and the science of research synthesis. *Nature* 555:175-182.
- Hágsater E. 2005. *Orchids of Mexico*. Productos Farmacéuticos, SA de CV.
- Halpern SL, Underwood N. 2006. Approaches for testing herbivore effects on plant population dynamics. *Journal of Applied Ecology* 43:922-929.
- Hansen TF, Stenseth NC, Henttonen H. 1999. Multiannual vole cycles and population regulation during long winters: an analysis of seasonal density dependence. *The American Naturalist* 154:129-139.
- Harper JL. 1977. *Population Biology of Plants*. Academic Press.
- Harris GM, Butler MJ, Stewart DR, Cain III JW. 2022. The abundance and persistence of Caprinae populations. *Scientific Reports* 12:13807.
- Hartemink N, Missov TI, Caswell H. 2017. Stochasticity, heterogeneity, and variance in

- longevity in human populations. *Theoretical Population Biology* 114:107-116.
- Hart-Fredeluces G, Ticktin T, Lake FK. 2021. Simulated Indigenous fire stewardship increases the population growth rate of an understorey herb. *Journal of Ecology* 109:1133-1147.
- Hastings A. 2001. Transient dynamics and persistence of ecological systems. *Ecology Letters* 4:215-220.
- Hastings A, Abbott KC, Cuddington K, Francis TB, Lai Y-C, Morozov A, Petrovskii S, Zeeman ML. 2021. Effects of stochasticity on the length and behaviour of ecological transients. *Journal of the Royal Society Interface* 18:20210257.
- Healy K, Ezard TH, Jones OR, Salguero-Gómez R, Buckley YM. 2019. Animal life history is shaped by the pace of life and the distribution of age-specific mortality and reproduction. *Nature Ecology & Evolution* 3:1217-1224.
- Heggerud C, Abbott K, Hastings A. 2023. *Transient dynamics*. Oxford University Press.
- Henneresse T, Wesselingh RA, Tyteca D. 2017. Effects of floral display, conspecific density and rewarding species on fruit set in the deceptive orchid *Orchis militaris* (Orchidaceae). *Plant Ecology and Evolution* 150:279-292.
- Hens H, Pakanen V-M, Jäkäläniemi A, Tuomi J, Kvist L. 2017. Low population viability in small endangered orchid populations: Genetic variation, seedling recruitment and stochasticity. *Biological Conservation* 210:174-183.
- Hernández-Apolinar M. 1992. Dinámica poblacional de *Laelia speciosa* (HBK) Schltr. (Orchidaceae). mathesis Thesis. Mexico City, Mexico: Facultad de Ciencias, UNAM.
- Hernández-Apolinar M, Gutiérrez-Paredes C, Luis P-GJ. 2012. Biología reproductiva de tres orquídeas mexicanas con distinto estatus de riesgo. :31-41.
- Hernández-Apolinar M, Gutiérrez-Paredes C, Sánchez-Gallen I, Aguirre E, Pérez-García E. 2012. Ecological aspects of *Cypripedium irapeanum* La Llave & Lex., an endangered Mexican orchid species. *SOA Newsletter* 13:1-5.
- Hernández CM, Daalen SF van, Caswell H, Neubert MG, Gribble KE. 2020. A demographic and evolutionary analysis of maternal effect senescence. *Proceedings of the National Academy of Sciences* 117:16431-16437.
- Hernández CM, Ellner SP, Adler PB, Hooker G, Snyder RE. 2023. An exact version of Life Table Response Experiment analysis, and the R package exactLTRE. *Methods in Ecology and Evolution* 14:939-951.
- Hernández CM, Ellner SP, Snyder RE, Hooker G. 2024. The natural history of luck: A synthesis study of structured population models. *Ecology Letters* 27:e14390.
- Herrera CM. 2009. *Multiplicity in unity: plant subindividual variation and interactions with animals*. University of Chicago Press.
- Hiernaux P, Diarra L, Trichon V, Mougin E, Soumaguel N, Baup F. 2009. Woody plant population dynamics in response to climate changes from 1984 to 2006 in Sahel (Gourma, Mali). *Journal of Hydrology* 375:103-113.
- Hietz P. 1997. Population dynamics of epiphytes in a Mexican humid montane forest. *Journal of Ecology*:767-775.
- Hinrichsen RA. 2025. Solving three core challenges in transient dynamics analysis of matrix population models. *bioRxiv*:2025-05.
- Hinsley A, De Boer HJ, Fay MF, Gale SW, Gardiner LM, Gunasekara RS, Kumar P, Masters S, Metusala D, Roberts DL, others. 2018. A review of the trade in orchids and its

- implications for conservation. *Botanical Journal of the Linnean Society* 186:435-455.
- Hodgson DJ, Townley S. 2004. Methodological insight: linking management changes to population dynamic responses: the transfer function of a projection matrix perturbation. *Journal of Applied Ecology* 41:1155-1161.
- Holttum RE. 1995. Growth-habitat of Monocotyledons-variations on a theme. *Phytomorphology* 5:399-413.
- Horvitz C, Schemske DW, Caswell H. 1997. The relative «importance» of life-history stages to population growth: prospective and retrospective analyses. In: *Structured-population models in marine, terrestrial, and freshwater systems*. Springer, 247-271.
- Hunter CM, Caswell H. 2005. Selective harvest of sooty shearwater chicks: effects on population dynamics and sustainability. *Journal of Animal Ecology*:589-600.
- Hunter CM, Caswell H, Runge MC, Regehr EV, Amstrup SC, Stirling I. 2010. Climate change threatens polar bear populations: a stochastic demographic analysis. *Ecology* 91:2883-2897.
- Hutchings MJ. 1987. The population biology of the early spider orchid, *Ophrys sphegodes* Mill. II. Temporal patterns in behaviour. *The Journal of Ecology*:729-742.
- Hutchings M. 2010. The population biology of the early spider orchid *Ophrys sphegodes* Mill. III. Demography over three decades. *Journal of Ecology* 98:867-878.
- Iralu V, Barbhuyan HSA, Upadhaya K. 2019. Ecology of seed germination in threatened trees: a review. *Energy, Ecology and Environment* 4:189-210.
- Iriondo Alegria J, Albert Gamboa M, Gimenez Benavides L, Dominguez Lozano F, Escudero Alcantara A. 2009. *Poblaciones en Peligro: Viabilidad Demográfica de la flora vascular amenazada de Espana*. Dirección General de Medio Ambiente y Política Forestal.
- Jacquemyn H, Brys R, Davison R, Tuljapurkar S, Jongejans E. 2012. Stochastic LTRE analysis of the effects of herbivory on the population dynamics of a perennial grassland herb. *Oikos* 121:211-218.
- Jacquemyn H, Brys R, Jongejans E. 2010. Seed limitation restricts population growth in shaded populations of a perennial woodland orchid. *Ecology* 91:119-129.
- Jäkäläniemi A, Crone EE, Närhi P, Tuomi J. 2011. Orchids do not pay costs at emergence for prolonged dormancy. *Ecology* 92:1538-1543.
- James TD, Salguero-Gómez R, Jones OR, Childs DZ, Beckerman AP. 2021. Bridging gaps in demographic analysis with phylogenetic imputation. *Conservation Biology* 35:1210-1221.
- Janovský Z, Herben T, Klimešová J. 2017. Accounting for clonality in comparative plant demography—growth or reproduction? *Folia Geobotanica* 52:433-442.
- Jansen M, Anten NP, Bongers F, Martínez-Ramos M, Zuidema PA. 2018. Towards smarter harvesting from natural palm populations by sparing the individuals that contribute most to population growth or productivity. *Journal of Applied Ecology* 55:1682-1691.
- Jelbert K, Buss D, McDonald J, Townley S, Franco M, Stott I, Jones O, Salguero-Gómez R, Buckley Y, Knight T, others. 2019. Demographic amplification is a predictor of invasiveness among plants. *Nature Communications* 10:5602.
- Jenouvrier S, Aubry LM, Barbraud C, Weimerskirch H, Caswell H. 2018. Interacting effects of unobserved heterogeneity and individual stochasticity in the life history of the southern fulmar. *Journal of Animal Ecology* 87:212-222.
- Jenouvrier S, Caswell H, Barbraud C, Weimerskirch H. 2010. Mating behavior, population growth, and the operational sex ratio: a periodic two-sex model approach. *The American*

- Naturalist* 175:739-752.
- Jenouvrier S, Holland M, Stroeve J, Barbraud C, Weimerskirch H, Serreze M, Caswell H. 2012. Effects of climate change on an emperor penguin population: analysis of coupled demographic and climate models. *Global Change Biology* 18:2756-2770.
- Jenouvrier S, Long MC, Coste CF, Holland M, Gamelon M, Yoccoz NG, Sæther B-E. 2022. Detecting climate signals in populations across life histories. *Global Change Biology* 28:2236-2258.
- Jepson J. 2000. The tree climber's companion. *Beaver Tree Pub., Longville, MN USA*.
- Jersáková J, Johnson SD, Kindlmann P. 2006. Mechanisms and evolution of deceptive pollination in orchids. *Biological Reviews* 81:219-235.
- Jersáková J, Malinová T. 2007. Spatial aspects of seed dispersal and seedling recruitment in orchids. *New Phytologist*:237-241.
- Jia P, Lu L, Chen X, Chen J, Guo L, Yu X, Liu Q. 2016. A climate-driven mechanistic population model of *Aedes albopictus* with diapause. *Parasites & vectors* 9:1-15.
- Jiang S, Jaggi H, Zuo W, Oli MK, Gaillard J-M, Tuljapurkar S. 2020. Life-history dynamics: damping time, demographic dispersion and generation time. *bioRxiv*:2020-12.
- Jiménez-Valdés M, Godínez-Alvarez H, Caballero J, Lira R. 2010. Population dynamics of *Agave marmorata* Roezl. under two contrasting management systems in Central Mexico. *Economic Botany* 64:149-160.
- Johansson D. 1974. Ecology of vascular epiphytes in West African rain forest. Thesis doctoral Thesis. Sv. växtgeografiska sällsk.
- Johnson S, Bond W. 1997. Evidence for widespread pollen limitation of fruiting success in Cape wildflowers. *Oecologia* 109:530-534.
- Johnson SD, Edwards T. 2000. The structure and function of orchid pollinaria. *Plant Systematics and Evolution* 222:243-269.
- Johnson DJ, Needham J, Xu C, Massoud EC, Davies SJ, Anderson-Teixeira KJ, Bunyavejchewin S, Chambers JQ, Chang-Yang C-H, Chiang J-M, others. 2018. Climate sensitive size-dependent survival in tropical trees. *Nature Ecology & Evolution* 2:1436-1442.
- Jones O, Barks P, Stott I, James T, Levin S, Petry W, Capdevila P, Che-Castaldo J, Jackson J, Römer G, others. 2022. Rcompadre and Rage—Two R packages to facilitate the use of the COMPADRE and COMADRE databases and calculation of life-history traits from matrix population models. *Methods in Ecology and Evolution* 13:770-781.
- Jones FA, Hubbell SP. 2006. Demographic spatial genetic structure of the Neotropical tree, *Jacaranda copaia*. *Molecular Ecology* 15:3205-3217.
- Jones OR, Scheuerlein A, Salguero-Gómez R, Camarda CG, Schaible R, Casper BB, Dahlgren JP, Ehrlén J, García MB, Menges ES, others. 2014. Diversity of ageing across the tree of life. *Nature* 505:169-173.
- Jops K, Dalling JW, O'Dwyer JP. 2025. Life history is a key driver of temporal fluctuations in tropical tree abundances. *Proceedings of the National Academy of Sciences* 122:e2422348122.
- Jouvet L, Rodríguez-Rojas A, Steiner UK. 2018. Demographic variability and heterogeneity among individuals within and among clonal bacteria strains. *Oikos* 127:728-737.
- Kendall BE, Fujiwara M, Diaz-Lopez J, Schneider S, Voigt J, Wiesner S. 2019. Persistent problems in the construction of matrix population models. *Ecological modelling* 406:33-

43.

- Kéry M, Gregg KB. 2004. Demographic analysis of dormancy and survival in the terrestrial orchid *Cypripedium reginae*. *Journal of Ecology* 92:686-695.
- Keyfitz N. 1968. Changing vital rates y age distributions. *Population Studies* 22:235-251.
- Kim Y, Kang K-W, Kim K-J. 2016. Restoration of endangered orchid species, *Dendrobium moniliforme* (L.) Sw.(Orchidaceae) in Korea. *Korean Journal of Plant Taxonomy* 46:256-266.
- Kindlmann JHW Pavel Willems. 2002. *Trends and fluctuations and underlying mechanisms in terrestrial orchid populations*. Leiden, The Netherlands: Backhuys Publishers.
- Kindlmann P, Meléndez-Ackerman EJ, Tremblay R. 2014. Disobedient epiphytes: colonization and extinction rates in a metapopulation of *Lepanthes rupestris* (Orchidaceae) contradict theoretical predictions based on patch connectivity. *Botanical Journal of the Linnean Society* 175:598-606.
- Klepac P, Caswell H. 2011. The stage-structured epidemic: linking disease and demography with a multi-state matrix approach model. *Theoretical Ecology* 4:301-319.
- Koons DN, Holmes RR, Grand JB. 2007. Population inertia and its sensitivity to changes in vital rates and population structure. *Ecology* 88:2857-2867.
- Koons DN, Iles DT, Stott I. 2021. Transient analyses of population dynamics using matrix projection models. *Demographic Methods across the Tree of Life (Oxford University Press Oxford, 2021)* p:197-212.
- Koop AL, Horvitz CC. 2005. Projection matrix analysis of the demography of an invasive, nonnative shrub (*Ardisia elliptica*). *Ecology* 86:2661-2672.
- Kroon H de, Plaisier A, Groenendaal J van, Caswell H. 1986. Elasticity: the relative contribution of demographic parameters to population growth rate. *Ecology* 67:1427-1431.
- Kull T. 1995. Genet and ramet dynamics of *Cypripedium calceolus* in different habitats. *Abstracta Botanica*:95-104.
- Larson RJ. 1992. Population dynamics of *Encyclia tampensis* in Florida. *Selbyana*:50-56.
- Larson SS, Batson WT. 1978. The vegetation of vertical rock faces in Pickens and Greenville counties, South Carolina. *Castanea*:255-260.
- Lawler RR, Caswell H, Richard AF, Ratsirarson J, Dewar RE, Schwartz M. 2009. Demography of Verreaux's sifaka in a stochastic rainfall environment. *Oecologia* 161:491-504.
- Lebreton J-D, Burnham KP, Clobert J, Anderson DR. 1992. Modeling survival and testing biological hypotheses using marked animals: a unified approach with case studies. *Ecological Monographs* 62:67-118.
- Lefkovich LP. 1965. The study of population growth in organisms grouped by stages. *Biometrics*:1-18.
- Leslie PH. 1945. On the use of matrices in certain population mathematics. *Biometrika* 33:183-212.
- Levin L, Caswell H, Bridges T, DiBacco C, Cabrera D, Plaia G. 1996. Demographic responses of estuarine polychaetes to pollutants: life table response experiments. *Ecological Applications* 6:1295-1313.
- Levin SC, Evers S, Potter T, Guerrero MP, Childs DZ, Compagnoni A, Knight TM, Salguero-Gómez R. 2022. Rpadrino: An R package to access and use PADRINO, an open access

- database of Integral Projection Models. *Methods in Ecology and Evolution* 13:1923-1929.
- Light MH, MacConaill M. 1998. Factors affecting germinable seed yield in *Cypripedium calceolus* var. *pubescens* (Willd.) Correll and *Epipactis helleborine* (L.) Crantz (Orchidaceae). *Botanical Journal of the Linnean Society* 126:3-26.
- Light MH, MacConaill M. 2006. Appearance and disappearance of a weedy orchid, *Epipactis helleborine*. *Folia Geobotanica* 41:77-93.
- Li P, Huang B, Pemberton R, Luo Y, Cheng J. 2011. Floral display influences male and female reproductive success of the deceptive orchid *Phaius delavayi*. *Plant Systematics and Evolution* 296:21-27.
- Li W-H, Ju Y-R, Liao C-M, Liao V-H-C. 2014. Assessment of selenium toxicity on the life cycle of *Caenorhabditis elegans*. *Ecotoxicology* 23:1245-1253.
- Lindenmayer DB, Possingham HP. 1996. Ranking conservation and timber management options for leadbeater's possum in southeastern Australia using population viability analysis. *Conservation Biology* 10:235-251.
- Lind H, Franzén M, Pettersson B, Anders Nilsson L. 2007. Metapopulation pollination in the deceptive orchid *Anacamptis pyramidalis*. *Nordic Journal of Botany* 25:176-182.
- Li S-L, Ramula S. 2015. Demographic strategies of plant invaders in temporally varying environments. *Population ecology* 57:373-380.
- Llabrés Fernández G. Estudio de la biología reproductiva de l'orquídia *Anacamptis coriophora*.
- Lopes EJM, Vilela B, Freitas Brito T de, Socorro Araújo de Moura T do, Nemésio A, Maués MM, Krug C, Martins MB, Paiva Silva D de, Dobrovolski R, others. 2022. Local abundance of neotropical orchid bees in Amazon forests not related to large-scale climate suitability. *Insect Conservation and Diversity* 15:693-703.
- Ludwig D. 1999. Is it meaningful to estimate a probability of extinction? *Ecology* 80:298-310.
- Luzuriaga AL, Escudero A. 2008. What determines emergence and net recruitment in an early succession plant community? Disentangling biotic and abiotic effects. *Journal of Vegetation Science* 19:445-456.
- Maldonado Flores C. 2005. Patrón de crecimiento y reproducción de la orquídea *Epidendrum xanthinum* Lindl. en la Cordillera Occidental, Valle del Cauca. mathesis Thesis. Instituto Politecnico Nacional, Santa Cruz Xoxocotlán, Oaxaca.
- Maldonado Flores C, others. 2006. Patrón de distribución espacial y dinámica poblacional de *Oncidium crista-galli*, una especie de orquídea epífita de Chiapas.
- Mandle L, Ticktin T, Zuidema PA. 2015. Resilience of palm populations to disturbance is determined by interactive effects of fire, herbivory and harvest. *Journal of Ecology* 103:1032-1043.
- Mangalam M, Kelty-Stephen DG. 2022. Ergodic descriptors of non-ergodic stochastic processes. *Journal of the Royal Society Interface* 19:20220095.
- Martínez-Villegas JA, Pisanty I, Martorell C, Hernández-Apolinar M, Valverde T, Granados-Hernández LA, Rodríguez-Sánchez M, Zúñiga-Vega JJ. 2024. Estimation of demographic parameters of some plant species accounting for imperfect detection probabilities. *Population Ecology* 66:22-38.
- McDonald JL, Stott I, Townley S, Hodgson DJ. 2016. Transients drive the demographic dynamics of plant populations in variable environments. *Journal of Ecology* 104:306-314.
- McGraw JB, Caswell H. 1996. Estimation of individual fitness from life-history data. *The*

- American Naturalist* 147:47-64.
- McMahon SM, Metcalf C. 2008. Transient sensitivities of non-indigenous shrub species indicate complicated invasion dynamics. *Biological Invasions* 10:833-846.
- Mehrhoff LA. 1989. The dynamics of declining populations of an endangered orchid, *Isotria medeoloides*. *Ecology* 70:783-786.
- Meléndez-Ackerman EJ, Ackerman JD, Rodríguez-Robles JA. 2000. Reproduction in an Orchid Can Be Resource-Limited over its Lifetime. *Biotropica* 32:282-290.
- Miller TE, Compagnoni A. 2022. Two-sex demography, sexual niche differentiation, and the geographic range limits of Texas bluegrass (*Poa arachnifera*). *The American Naturalist* 200:17-31.
- Miriti MN, Joseph Wright S, Howe HF. 2001. The effects of neighbors on the demography of a dominant desert shrub (*Ambrosia dumosa*). *Ecological Monographs* 71:491-509.
- Mitchell-Olds T. 1987. Analysis of local variation in plant size. *Ecology* 68:82-87.
- Moles AT, Westoby M. 2006. Seed size and plant strategy across the whole life cycle. *Oikos* 113:91-105.
- Mondragón D. 2000. Dinámica poblacional de *Tillandsia brachycaulos* Schltdl. en el Parque Nacional de Dzibichaltún, Yucatán. Tesis doctoral Thesis. Centro de Investigación Científica de Yucatán.
- Mondragón D. 2009. Population viability analysis for Guarianthe aurantiaca, an ornamental epiphytic orchid harvested in Southeast México. *Plant species biology* 24:35-41.
- Mondragón Chaparro DM, Valverde Valdés MT, Hernández Apolinar M, others. 2015. Population ecology of epiphytic angiosperms: a review. *Tropical Ecology* 56:01-39.
- Mondragón D, Elliott DD. 2013. The population dynamics of epiphytic orchids: a review and methodological guide. *Treetops at Risk: Challenges of Global Canopy Ecology and Conservation*:377-385.
- Mondragón D, Maldonado C, Aguilar-Santelises R. 2007. Life history and demography of a twig epiphyte: a case study of *Erycina crista-galli* (Orchidaceae). *Selbyana*:137-144.
- Morris WF, Doak DF. 2002. *Quantitative conservation biology: theory and practice of population viability analysis*. Sinauer Associates, Inc. Publishers.
- Morris WF, Ehrlén J, Dahlgren JP, Loomis AK, Louthan AM. 2020. Biotic and anthropogenic forces rival climatic/abiotic factors in determining global plant population growth and fitness. *Proceedings of the National Academy of Sciences* 117:1107-1112.
- Namkoong G, Roberds J. 1974. Extinction probabilities and the changing age structure of redwood forests. *The American Naturalist* 108:355-368.
- Nantel P, Gagnon D, Nault A. 1996. Population viability analysis of American ginseng and wild leek harvested in stochastic environments. *Conservation Biology* 10:608-621.
- Neiland MRM, Wilcock CC. 1998. Fruit set, nectar reward, and rarity in the Orchidaceae. *American Journal of Botany* 85:1657-1671.
- Neubert MG, Caswell H. 2000. Demography and dispersal: calculation and sensitivity analysis of invasion speed for structured populations. *Ecology* 81:1613-1628.
- Nguyen V, Buckley YM, Salguero-Gómez R, Wardle GM. 2019. Consequences of neglecting cryptic life stages from demographic models. *Ecological Modelling* 408:108723.
- Nicolè F, Brzosko E, Till-Bottraud I. 2005. Population viability analysis of *Cypripedium calceolus* in a protected area: longevity, stability and persistence. *Journal of Ecology* 93:716-726.

- Nicotra AB, Atkin OK, Bonser SP, Davidson AM, Finnegan EJ, Mathesius U, Poot P, Purugganan MD, Richards CL, Valladares F, others. 2010. Plant phenotypic plasticity in a changing climate. *Trends in plant science* 15:684-692.
- Nordbakken J-F, Rydgren K, Økland R. 2004. Demography and population dynamics of *Drosera anglica* and *D. rotundifolia*. *Journal of Ecology* 92:110-121.
- Nordstrom SW, Dykstra AB, Wagenius S. 2021. Fires slow population declines of a long-lived prairie plant through multiple vital rates. *Oecologia* 196:679-691.
- Novoa A, Le Roux JJ, Robertson MP, Wilson JR, Richardson DM. 2015. Introduced and invasive cactus species: a global review. *AoB Plants* 7:plu078.
- Nunney L, Campbell KA. 1993. Assessing minimum viable population size: demography meets population genetics. *Trends in Ecology & Evolution* 8:234-239.
- Oli MK, Slade NA, Dobson FS. 2001. Effect of density reduction on Uinta ground squirrels: analysis of life table response experiments. *Ecology* 82:1921-1929.
- Omeyer L, Stokes K, Beton D, Çiçek B, Davey S, Fuller W, Godley B, Sherley R, Snape R, Broderick A. 2021. Investigating differences in population recovery rates of two sympatrically nesting sea turtle species. *Animal Conservation* 24:832-846.
- Opedal ØH, Falahati-Anbaran M, Albertsen E, Armbruster WS, Pérez-Barrales R, Stenøien HK, Pélabon C. 2017. Euglossine bees mediate only limited long-distance gene flow in a tropical vine. *New Phytologist* 213:1898-1908.
- Orive ME. 1993. Effective population size in organisms with complex life-histories. *Theoretical Population Biology* 44:316-340.
- Ospina-Calderón N, Tremblay R, Torres A, Flanagan N. 2023. The effect of habitat transformation on a twig epiphytic orchid: Evidence from population dynamics. *Frontiers in Ecology and Evolution* 11:1135316.
- Paiva Neto VB de, Torrezan M de A, Silva MAV da, Padilha DRC, Borel JC, Zuffo-Borges MCR. 2022. Autopolinização de *Cynoches haagii*, orquídea nativa do Cerrado Brasileiro, resulta em plântulas fenotipicamente albinas. *Ornamental Horticulture* 28:85-91.
- Paniw M, Maag N, Cozzi G, Clutton-Brock T, Ozgul A. 2019. Life history responses of meerkats to seasonal changes in extreme environments. *Science* 363:631-635.
- Paniw M, Quintana-Ascencio PF, Ojeda F, Salguero-Gómez R. 2017. Accounting for uncertainty in dormant life stages in stochastic demographic models. *Oikos* 126:900-909.
- Pantone D, Baker J, Jordan P. 1992. Path Analysis of Red Rice (*Oryza sativa* L.) Competition by Cultivated Rice. *Weed Science* 40:313-319.
- Pardini EA, Drake JM, Chase JM, Knight TM. 2009. Complex population dynamics and control of the invasive biennial *Alliaria petiolata* (garlic mustard). *Ecological Applications* 19:387-397.
- Parés-Ramos IK, Gould WA, Aide TM. 2008. Agricultural abandonment, suburban growth, and forest expansion in Puerto Rico between 1991 and 2000. *Ecology and Society* 13.
- Pascarella JB, Horvitz CC. 1998. Hurricane disturbance and the population dynamics of a tropical understory shrub: megamatrix elasticity analysis. *Ecology* 79:547-563.
- Peakall R. 1989. A new technique for monitoring pollen flow in orchids. *Oecologia* 79:361-365.
- Pearl R, Miner JR, Parker SL. 1927. Experimental studies on the duration of life. XI. Density of population and life duration in *Drosophila*. *The American Naturalist* 61:289-318.
- Pellegrino G, Bellusci F. 2014. Effects of human disturbance on reproductive success

- and population viability of *Serapias cordigera* (Orchidaceae). *Botanical Journal of the Linnean Society* 176:408-420.
- Peña-Domene M de la, Martínez-Garza C, Howe HF. 2013. Early recruitment dynamics in tropical restoration. *Ecological Applications* 23:1124-1134.
- Petrolli R, Zinger L, Perez-Lamarque B, Collobert G, Griveau C, Paillet T, Selosse M-A, Martos F. 2022. Spatial turnover of fungi and partner choice shape mycorrhizal networks in epiphytic orchids. *Journal of Ecology* 110:2568-2584.
- Pfister CA. 1998. Patterns of variance in stage-structured populations: evolutionary predictions and ecological implications. *Proceedings of the National Academy of Sciences* 95:213-218.
- Plard F, Fay R, Kéry M, Cohas A, Schaub M. 2019. Integrated population models: powerful methods to embed individual processes in population dynamics models. *Ecology* 100:e02715.
- Pokorny T, Loose D, Dyker G, Quezada-Euán JJG, Eltz T. 2015. Dispersal ability of male orchid bees and direct evidence for long-range flights. *Apidologie* 46:224-237.
- Postma JA, Hecht VL, Hikosaka K, Nord EA, Pons TL, Poorter H. 2021. Dividing the pie: A quantitative review on plant density responses. *Plant, Cell & Environment* 44:1072-1094.
- Powers SM, Hampton SE. 2019. Open science, reproducibility, and transparency in ecology. *Ecological Applications* 29:e01822.
- Primack R, Stacy E. 1998. Cost of reproduction in the pink lady's slipper orchid (*Cypripedium acaule*, Orchidaceae): an eleven-year experimental study of three populations. *American Journal of Botany* 85:1672-1679.
- Proctor HC, Harder LD. 1994. Pollen load, capsule weight, and seed production in three orchid species. *Canadian Journal of Botany* 72:249-255.
- Rademaker M, Leeuwen A van, Smallegange IM. 2024. Why we cannot always expect life history strategies to directly inform on sensitivity to environmental change. *Journal of Animal Ecology* 93:348-366.
- Ramírez Martínez A, others. 2021. Host tree effect on demography and phenology of epiphytic species.
- Ramula S, Knight TM, Burns JH, Buckley YM. 2008. General guidelines for invasive plant management based on comparative demography of invasive and native plant populations. *Journal of Applied Ecology* 45:1124-1133.
- Rasmussen HN. 1995. *Terrestrial orchids: from seed to mycotrophic plant*. Cambridge University Press.
- Rasmussen FN. 1999. 12 The development of orchid classification. *Genera Orchidacearum: Volume 1: Apostasioideae and Cypripedioideae* 1:12.
- Rasmussen HN. 2011. Methods of studying field germination and seedling physiology: present potential and drawbacks. *European Journal of Environmental Sciences* 1.
- Rasmussen HN, Dixon KW, Jersáková J, Těšitelová T. 2015. Germination and seedling establishment in orchids: a complex of requirements. *Annals of Botany* 116:391-402.
- Rasmussen HN, Whigham DF. 1993. Seed ecology of dust seeds in situ: a new study technique and its application in terrestrial orchids. *American Journal of Botany* 80:1374-1378.
- Raventós J, García-González A, Riverón-Giró FB, Damon A. 2018. Comparison of transient and asymptotic perturbation analyses of three epiphytic orchid species growing in coffee

- plantations in Mexico: effect on conservation decisions. *Plant Ecology & Diversity* 11:133-145.
- Raventós J, González E, Mújica E, Bonet A. 2015. Transient population dynamics of two epiphytic orchid species after Hurricane Ivan: implications for management. *Biotropica* 47:441-448. DOI: [10.1111/btp.12231](https://doi.org/10.1111/btp.12231).
- Raventós J, Mújica E, Gonzalez E, Bonet A, Ortega-Larrocea MP. 2021. The effects of hurricanes on the stochastic population growth of the endemic epiphytic orchid *Broughtonia cubensis* living in Cuba. *Population Ecology* 63:302-312.
- R JO, Barks P, Stott IM, James TD, Levin SC, Petry WK, Capdevila P, Che-Castaldo J, Jackson J, Römer G, Schuette C, Thomas CC, Salguero-Gómez R. 2022. Rcompadre and Rage - two R packages to facilitate the use of the COMPADRE and COMADRE databases and calculation of life history traits from matrix population models. *Methods in Ecology and Evolution* 13:770-781. DOI: [10.1111/2041-210X.13792](https://doi.org/10.1111/2041-210X.13792).
- Regehr EV, Hunter CM, Caswell H, Amstrup SC, Stirling I. 2010. Survival and breeding of polar bears in the southern Beaufort Sea in relation to sea ice. *Journal of Animal Ecology* 79:117-127.
- Reichman OJ, Jones MB, Schildhauer MP. 2011. Challenges and opportunities of open data in ecology. *Science* 331:703-705.
- Riecke TV, Williams PJ, Behnke TL, Gibson D, Leach AG, Sedinger BS, Street PA, Sedinger JS. 2019. Integrated population models: model assumptions and inference. *Methods in Ecology and Evolution* 10:1072-1082.
- Riverón-Giró FB, Raventós J, Damon A, García-González A, Mújica E. 2019. Spatio-temporal dynamics of the invasive orchid *Oeceoclades maculata* (Orchidaceae), in four different habitats in southeast Chiapas, Mexico. *Biological Invasions* 21:1905-1919.
- Rodríguez-Robles JA, Ackerman JD, Meléndez E. 1990. Host distribution and hurricane damage to an orchid population at Toro Negro Forest, Puerto Rico. *Caribbean Journal of Science* 26:163-164.
- Römer G, Dahlgren JP, Salguero-Gómez R, Stott IM, Jones OR. 2023. Plant demographic knowledge is biased towards short-term studies of temperate-region herbaceous perennials. *Oikos*:e10250.
- Roth G, Caswell H. 2016. Hyperstate matrix models: extending demographic state spaces to higher dimensions. *Methods in Ecology and Evolution* 7:1438-1450.
- Roux PC le, McGeoch MA. 2004. The use of size as an estimator of age in the subantarctic cushion plant, *Azorella selago* (Apiaceae). *Arctic, Antarctic, and Alpine Research* 36:509-517.
- Saatkamp A, Poschlod P, Venable DL. 2014. 11 The Functional Role of Soil Seed Banks in Natural Communities.
- Sæther B-E, Coulson T, Grøtan V, Engen S, Altwegg R, Armitage KB, Barbraud C, Becker PH, Blumstein DT, Dobson FS, others. 2013. How life history influences population dynamics in fluctuating environments. *The American Naturalist* 182:743-759.
- Sagar G. 1985. Profile of John L. Harper. *Studies of plant demography: a festschrift for John L. Harper*:xix-xxv.
- Salguero-Gómez R, Jackson J, Gascoigne SJ. 2021. Four key challenges in the open-data revolution. *Journal of Animal Ecology* 90:2000-2004.
- Salguero-Gómez R, Jones OR, Archer CR, Bein C, Buhr H de, Farack C, Gottschalk F,

- Hartmann A, Henning A, Hoppe G, others. 2016. COMADRE: a global data base of animal demography. *Journal of Animal Ecology* 85:371-384.
- Salguero-Gómez R, Jones OR, Archer CR, Buckley YM, Che-Castaldo J, Caswell H, Hodgson D, Scheuerlein A, Conde DA, Brinks E, others. 2015. The compadre Plant Matrix Database: an open online repository for plant demography. *Journal of Ecology* 103:202-218.
- Salguero-Gomez R, Jones OR, Jongejans E, Blomberg SP, Hodgson DJ, Mbeau-Ache C, Zuidema PA, Kroon H de, Buckley YM. 2017. Fast-slow continuum and reproductive strategies structure plant life-history variation worldwide (vol 113, pg 230, 2015). *Proceedings of the National Academy of Sciences of the United States of America* 114:E9753.
- Salguero-Gómez R, Plotkin JB. 2010. Matrix dimensions bias demographic inferences: implications for comparative plant demography. *The American Naturalist* 176:710-722.
- Sampaio MC, Picó FX, Scarano FR. 2005. Ramet demography of a nurse bromeliad in Brazilian restingas. *American Journal of Botany* 92:674-681.
- Sarukhan J, Gadgil M. 1974. Studies on plant demography: *Ranunculus repens* L., *R. bulbosus* L. and *R. acris* L.: III. a mathematical model incorporating multiple modes of reproduction. *The Journal of Ecology*:921-936.
- Schaub M, Kéry M. 2021. *Integrated population models: Theory and ecological applications with R and JAGS*. Academic Press.
- Schefferson R. 2002. Dormancy and survival in a rare terrestrial orchid. In: Kindlmann P, JH W, Whigham D eds. *Trends and Fluctuations and Underlying Mechanisms in Terrestrial Orchid Populations*. Backhuys Publishers, Leiden, The Netherlands, 53-64.
- Schmidt IB, Mandle L, Ticktin T, Gaoue OG. 2011. What do matrix population models reveal about the sustainability of non-timber forest product harvest? *Journal of Applied Ecology* 48:815-826.
- Schödelbauerová I, Tremblay R, Kindlmann P. 2010. Prediction vs. reality: Can a PVA model predict population persistence 13 years later? *Biodiversity and Conservation* 19:637-650. DOI: [10.1007/s10531-009-9724-1](https://doi.org/10.1007/s10531-009-9724-1).
- Schoot R van de, Depaoli S, King R, Kramer B, Märten K, Tadesse MG, Vannucci M, Gelman A, Veen D, Willemsen J, others. 2021. Bayesian statistics and modelling. *Nature Reviews Methods Primers* 1:1.
- Scopece G, Juillet N, Lexer C, Cozzolino S. 2017. Fluctuating selection across years and phenotypic variation in food-deceptive orchids. *PeerJ* 5:e3704.
- Semmens BX, Semmens DJ, Thogmartin WE, Wiederholt R, López-Hoffman L, Diffendorfer JE, Pleasants JM, Oberhauser KS, Taylor OR. 2016. Quasi-extinction risk and population targets for the Eastern, migratory population of monarch butterflies (*Danaus plexippus*). *Scientific reports* 6:23265.
- Shaffer ML. 1981. Minimum population sizes for species conservation. *BioScience* 31:131-134.
- Shaffer ML, Samson FB. 1985. Population size and extinction: a note on determining critical population sizes. *The American Naturalist* 125:144-152.
- Shao S-C, Jacquemyn H, Selosse M-A. 2024. Improved use of mycorrhizal fungi for upscaling of orchid cultivation and population restoration of endangered orchids in China. *Symbiosis*:1-10.

- Shefferson RP, Jacquemyn H, Kull T, Hutchings MJ. 2020. The demography of terrestrial orchids: life history, population dynamics and conservation. *Botanical Journal of the Linnean Society* 192:315-332.
- Shefferson RP, Kull T, Tali K. 2005. Adult whole-plant dormancy induced by stress in long-lived orchids. *Ecology* 86:3099-3104.
- Shefferson RP, Kull T, Tali K, Kellett KM. 2012. Linking vegetative dormancy to fitness in two long-lived herbaceous perennials. *Ecosphere* 3:1-19. DOI: [10.1890/ES11-00328.1](https://doi.org/10.1890/ES11-00328.1).
- Shefferson RP, Mizuta R, Hutchings MJ. 2017. Predicting evolution in response to climate change: the example of sprouting probability in three dormancy-prone orchid species. *Royal Society Open Science* 4:160647.
- Shefferson RP, Proper J, Beissinger SR, Simms EL. 2003. Life history trade-offs in a rare orchid: the costs of flowering, dormancy, and sprouting. *Ecology* 84:1199-1206.
- Shefferson RP, Sandercock BK, Proper J, Beissinger SR. 2001. Estimating dormancy and survival of a rare herbaceous perennial using mark-recapture models. *Ecology* 82:145-156.
- Shefferson RP, Tali K. 2007. Dormancy is associated with decreased adult survival in the burnt orchid, *Neotinea ustulata*. *Journal of Ecology* 95:217-225.
- Shefferson RP, Warren RJ, Pulliam HR. 2014. Life-history costs make perfect sprouting maladaptive in two herbaceous perennials. *Journal of Ecology* 102:1318-1328.
- Shyu E, Caswell H. 2016a. A demographic model for sex ratio evolution and the effects of sex-biased offspring costs. *Ecology and Evolution* 6:1470-1492.
- Shyu E, Caswell H. 2016b. Frequency-dependent two-sex models: A new approach to sex ratio evolution with multiple maternal conditions. *Ecology and Evolution* 6:6855-6879.
- Shyu E, Pardini EA, Knight TM, Caswell H. 2013. A seasonal, density-dependent model for the management of an invasive weed. *Ecological Applications* 23:1893-1905.
- Sieg CH, King RM. 1995. Influence of environmental factors and preliminary demographic analyses of a threatened orchid, *Platanthera praeclara*. *American Midland Naturalist*:307-323.
- Silvertown J, Charlesworth D. 2009. *Introduction to Plant Population Biology*. John Wiley & Sons.
- Silvertown J, Franco M, Menges E. 1996. Interpretation of elasticity matrices as an aid to the management of plant populations for conservation. *Conservation Biology* 10:591-597.
- Simmons C. 2016. Conservation genetics and population viability of the endangered swamp orchid *Phaius australis* (Orchidaceae) in a changing world. Tesis doctoral Thesis. Faculty of Science, Health, Education; Engineering, University of the Sunshine Coast, Queensland, Australia: University of the Sunshine Coast.
- Skalski JR, Ryding KE, Millspaugh J. 2010. *Wildlife demography: analysis of sex, age, and count data*. Elsevier.
- Sky H, Nick, Jackson J, Chege G, Gaymer J, Kimiti D, Mutisya S, Nakito S, Shultz S. 2022. Female reproductive skew exacerbates the extinction risk from poaching in the eastern black rhino. *Proceedings of the Royal Society B* 289:20220075.
- Sletvold N, Ågren J. 2011. Among-population variation in costs of reproduction in the long-lived orchid *Gymnadenia conopsea*: an experimental study. *Oecologia* 167:461-468.
- Sletvold N, Dahlgren JP, Øien D-I, Moen A, Ehrlén J. 2013. Climate warming alters effects of management on population viability of threatened species: results from a 30-year experimental study on a rare orchid. *Global Change Biology* 19:2729-2738.

- Sletvold N, Øien D-I, Moen A. 2010. Long-term influence of mowing on population dynamics in the rare orchid *Dactylorhiza lapponica*: the importance of recruitment and seed production. *Biological Conservation* 143:747-755.
- Smith ZF. 2006. Developping a Reintroduction Program for the Threatened Terrestrial Orchid *Diuris fragrantissima*. Tesis doctoral Thesis. School of Resource Management, Faculty of Land; Food Resources: The University of Melbourne.
- Smith M, Caswell H, Mettler-Cherry P. 2005. Stochastic flood and precipitation regimes and the population dynamics of a threatened floodplain plant. *Ecological Applications* 15:1036-1052.
- Snyder RE, Ellner SP. 2018. Pluck or luck: does trait variation or chance drive variation in lifetime reproductive success? *The American Naturalist* 191:E90-E107.
- Snyder RE, Ellner SP. 2022. Snared in an evil time: How age-dependent environmental and demographic variability contribute to variance in lifetime outcomes. *The American Naturalist* 200:E124-E140.
- Souza JM, Schmidt IB, Conceicao AA. 2018. How do fire and harvesting affect the population dynamics of a dominant endemic Velloziaceae species in campo rupestre? *Flora* 238:225-233.
- Steiner UK, Tuljapurkar S, Roach DA. 2021. Quantifying the effect of genetic, environmental and individual demographic stochastic variability for population dynamics in *Plantago lanceolata*. *Scientific Reports* 11:23174.
- Stott IM. 2012. *Modelling transient population dynamics and their role in ecology and evolution*. University of Exeter (United Kingdom).
- Stott I. 2016. Perturbation analysis of transient population dynamics using matrix projection models. *Methods in Ecology and Evolution* 7:666-678.
- Stott I, Franco M, Carslake D, Townley S, Hodgson D. 2010a. Boom or bust? A comparative analysis of transient population dynamics in plants. *Journal of Ecology* 98:302-311.
- Stott I, Hodgson DJ, Townley S. 2012b. popdemo: an R package for population demography using projection matrix analysis. *Methods in Ecology and Evolution* 3:797-802.
- Stott I, Hodgson DJ, Townley S. 2012a. Beyond sensitivity: nonlinear perturbation analysis of transient dynamics. *Methods in Ecology and Evolution* 3:673-684.
- Stott I, Hodgson D, Townley S. 2021. *popdemo: Demographic Modelling Using Projection Matrices*.
- Stott I, Townley S, Carslake D, Hodgson DJ. 2010b. On reducibility and ergodicity of population projection matrix models. *Methods in Ecology and Evolution* 1:242-252.
- Stott I, Townley S, Hodgson DJ. 2011. A framework for studying transient dynamics of population projection matrix models. *Ecology Letters* 14:959-970.
- Stubben CJ, Milligan BG. 2007. Estimating and Analyzing Demographic Models Using the popbio Package in R. *Journal of Statistical Software* 22. DOI: [10.18637/jss.v022.i11](https://doi.org/10.18637/jss.v022.i11).
- Subedi A, Kunwar B, Choi Y, Dai Y, Anel T van, Chaudhary RP, Boer HJ de, Gravendeel B. 2013. Collection and trade of wild-harvested orchids in Nepal. *Journal of Ethnobiology and Ethnomedicine* 9:1-10.
- Suetsugu K. 2015. Autonomous self-pollination and insect visitors in partially and fully mycoheterotrophic species of *Cymbidium* (Orchidaceae). *Journal of Plant Research* 128:115-125.
- Suetsugu K, Fukushima S. 2014. Pollination biology of the endangered orchid *Cypripedium*

- japonicum* in a fragmented forest of Japan. *Plant Species Biology* 29:294-299.
- Sun W, Yuan X, Liu Z-J, Lan S, Tsai W, Zou S-Q. 2019. Multivariate analysis reveals phenotypic diversity of *Euscaphis japonica* population. *PloS one* 14:e0219046.
- Tałałaj I, Ostrowiecka B, Włostowska E, Rutkowska A, Brzosko E. 2017. The ability of spontaneous autogamy in four orchid species: *Cephalanthera rubra*, *Neottia ovata*, *Gymnadenia conopsea*, and *Platanthera bifolia*. *Acta Biologica Cracoviensia. Series Botanica* 59.
- Talalaj I, Skierczynski M. 2015. Mechanism of spontaneous autogamy in the allogamous lepidopteran orchid *Gymnadenia conopsea* (L.) R. Br. (Orchidaceae). *Acta Biologica Cracoviensia. Series Botanica* 57.
- Tali K. 2002. Dynamics of *Orchis ustulata* populations in Estonia. In: Kindlmann P, JH W, Whigham D eds. *Trends and Fluctuations and Underlying Mechanisms in Terrestrial Orchid Populations*. Backhuys Publishers, Leiden, The Netherlands, 33-42.
- Tamm CO. 1948. Observations on reproductive and survival of some perennial herbs. *Botanish Notiser*:305-321.
- Tamm CO. 1956. Further observations on the survival and flowering of some perennial herbs, I. *Oikos* 7:273-292.
- Tamm CO. 1972. Survival and flowering of some perennial herbs. II. The behaviour of some orchids on permanent plots. *Oikos*:23-28.
- Teitel Z, Klimowski A, Campbell LG. 2016. Assessing the effects of hybridization and precipitation on invasive weed demography using strength of selection on vital rates. *BMC Evolutionary Biology* 16:1-13.
- Téllez V, Flores L. 2007. Orquídeas terrestres del Pedregal de San Ángel. *UNAM. México, DF*.
- Temple SA. 1977. Plant-animal mutualism: coevolution with dodo leads to near extinction of plant. *Science* 197:885-886.
- Tenhumberg B, Louda SM, Eckberg JO, Takahashi M. 2008. Monte Carlo analysis of parameter uncertainty in matrix models for the weed *Cirsium vulgare*. *Journal of Applied Ecology* 45:438-447.
- Tenhumberg B, Tyre AJ, Rebarber R. 2009. Model complexity affects transient population dynamics following a dispersal event: a case study with pea aphids. *Ecology* 90:1878-1890.
- Thorpe A, Stanley A, Kaye T, P L. 2011. *Population trends, demography, and the effects of environment and disturbance on Cypripedium fasciculatum in southern Oregon*. Institute for Applied Ecology, Corvallis, Oregon; USDI Bureau of Land Management, Medford District, Medford, Oregon.
- Ticktin T, Mondragón D, Lopez-Toledo L, Dutra-Elliott D, Aguirre-León E, Hernández-Apolinar M. 2020. Synthesis of wild orchid trade and demography provides new insight on conservation strategies. *Conservation Letters* 13:e12697.
- Ticktin T, Nantel P, Ramirez F, Johns T. 2002. Effects of variation on harvest limits for nontimber forest species in Mexico. *Conservation Biology* 16:691-705.
- Timsina B, Kindlmann P, Münzbergová Z, Rokaya MB. 2021. Six-year demographic study of the terrestrial orchid, *Crepidium acuminatum*: implications for conservation. *Frontiers in Ecology and Evolution* 9:676993.
- Tomowski M, Lozada-Gobilard S, Jeltsch F, Tiedemann R. 2023. Recruitment and migration

- patterns reveal a key role for seed banks in the meta-population dynamics of an aquatic plant. *Scientific Reports* 13:11269.
- Tonhasca JA, Albuquerque GS, Blackmer JL. 2003. Dispersal of euglossine bees between fragments of the Brazilian Atlantic Forest. *Journal of Tropical Ecology* 19:99-102.
- Torres AD, Cellini JM, Lencinas MV, Barrera MD, Soler R, Díaz Delgado R, Martínez Pastur G. 2015. Seed production and recruitment in primary and harvested *Nothofagus pumilio* forests: influence of regional climate and years after cuttings. *Forest Systems* 24.
- Traill LW, Bradshaw CJ, Brook BW. 2007. Minimum viable population size: a meta-analysis of 30 years of published estimates. *Biological Conservation* 139:159-166.
- Trapnell DW, Hamrick J, Nason JD. 2004. Three-dimensional fine-scale genetic structure of the neotropical epiphytic orchid, *Laelia rubescens*. *Molecular Ecology* 13:1111-1118.
- Tremblay RL. 1992. Trends in the pollination ecology of the Orchidaceae: evolution and systematics. *Canadian Journal of Botany* 70:642-650.
- Tremblay R. 1994. Frequency and consequences of multi-parental pollinations in a population of *Cypripedium calceolus* var. *pubescens* (Orchidaceae). *Lindleyana* 9:161-167.
- Tremblay R. 1997a. *Lepanthes caritensis*, an endangered orchid: no sex, no future? *Selbyana*:160-166.
- Tremblay R. 1997b. Distribution and Dispersion Patterns of Individuals in Nine Species of *Lepanthes* (Orchidaceae). *Biotropica* 29:38-45.
- Tremblay R. 2000. Plant longevity in four species of *Lepanthes* (Pleurothallidinae; Orchidaceae). *Lindleyana* 15:257-266.
- Tremblay R. 2006. The effect of flower position on male and female reproductive success in a deceptively pollinated tropical orchid. *Botanical Journal of the Linnean Society* 151:405-410.
- Tremblay R, Ackerman J. 2001. Gene flow and effective population size in *Lepanthes* (Orchidaceae): a case for genetic drift. *Biological Journal of the Linnean Society* 72:47-62. DOI: [10.1006/bj1.2000.0485](https://doi.org/10.1006/bj1.2000.0485).
- Tremblay R, Ackerman JD, Pérez M-E. 2010. Riding across the selection landscape: fitness consequences of annual variation in reproductive characteristics. *Philosophical Transactions of the Royal Society B: Biological Sciences* 365:491-498.
- Tremblay R, Ackerman J, Zimmerman J, Calvo R. 2005. Variation in sexual reproduction in orchids and its evolutionary consequences: a spasmodic journey to diversification. *Biological Journal of the Linnean Society* 84:1-54.
- Tremblay RL, Hutchings MJ. 2003. Population dynamics in orchid conservation: a review of analytical methods based on the rare species *Lepanthes eltoroensis*. In: *Orchid Conservation*. Kota Kinabalu: Natural History Publications (Borneo), 183-204.
- Tremblay R, McCarthy MA. 2014. Bayesian estimates of transition probabilities in seven small lithophytic orchid populations: maximizing data availability from many small samples. *PLoS One* 9:e102859.
- Tremblay, Meléndez-Ackerman E, Kapan D. 2006. Do epiphytic orchids behave as metapopulations? Evidence from colonization, extinction rates and asynchronous population dynamics. *Biological Conservation* 129:70-81.
- Tremblay R, Perez M, Larcombe M, Brown A, Quarmby J, Bickerton D, French G, Bould A. 2009b. Dormancy in *Caladenia*: A Bayesian approach to evaluating latency. *Australian Journal of Botany* 57:340-350.

- Tremblay R, Perez M-E, Larcombe M, Brown A, Quarmby J, Bickerton D, French G, Bould A. 2009a. Population dynamics of *Caladenia*: Bayesian estimates of transition and extinction probabilities. *Australian Journal of Botany* 57:351-360.
- Tremblay R, Raventos J, Ackerman J. 2015. When stable-stage equilibrium is unlikely: integrating transient population dynamics improves asymptotic methods. *Annals of Botany* 116:381-390. DOI: [10.1093/aob/mcv031](https://doi.org/10.1093/aob/mcv031).
- Tremblay RL, Tyre AJ, Pérez M-E, Ackerman JD. 2021. Population projections from holey matrices: Using prior information to estimate rare transition events. *Ecological Modelling* 447:109526.
- Tuljapurkar S. 1989. An uncertain life: demography in random environments. *Theoretical Population Biology* 35:227-294.
- Tuljapurkar S. 2013. *Population dynamics in variable environments*. Springer Science & Business Media.
- Tyre A, Tremblay R, Perez M. 2024. *raretrans: Bayesian Priors for Matrix Population Models*.
- Vallejo-Marín M, Dorken ME, Barrett SC. 2010. The ecological and evolutionary consequences of clonality for plant mating. *Annual Review of Ecology, Evolution, and Systematics* 41:193-213.
- Van Groenendael J, De Kroon H, Kalisz S, Tuljapurkar S. 1994. Loop analysis: evaluating life history pathways in population projection matrices. *Ecology* 75:2410-2415.
- Vanderklein D, Martinez-Vilalta J, Lee S, Mencuccini M. 2007. Plant size, not age, regulates growth and gas exchange in grafted Scots pine trees. *Tree physiology* 27:71-79.
- Vanhecke L. 1991. Population dynamics of *Dactylorhiza praetermissa* in relation to topography and inundation. *Population Ecology of Terrestrial Orchids*. SPB Academic Publishing, The Hague:15-32.
- Veran S, Beissinger SR. 2009. Demographic origins of skewed operational and adult sex ratios: perturbation analyses of two-sex models. *Ecology Letters* 12:129-143.
- Vindenes Y, Le Coeur C, Caswell H, others. 2021. Introduction to matrix population models. *Demographic Methods across the Tree of Life*:163-81.
- Vries C de, Caswell H. 2019. Selection in two-sex stage-structured populations: Genetics, demography, and polymorphism. *Theoretical Population Biology* 130:160-169.
- Waite S, Hutchings M. 1991. The effects of different management regimes on the population dynamics of *Ophrys sphegodes*: analysis and description using matrix models. In: Wells TCE, Willems JH eds. *Population ecology of terrestrial orchids*. SPB Academic Publishing bv, The Hague, 161-175.
- Weiner J, Stoll P, Muller-Landau H, Jasentuliyana A. 2001. The effects of density, spatial pattern, and competitive symmetry on size variation in simulated plant populations. *The American Naturalist* 158:438-450.
- Weiner J, Thomas SC. 1986. Size variability and competition in plant monocultures. *Oikos*:211-222.
- Wells T. 1991. Demographic and biological studies on *Ophrys apifera*: some results from a 10 year study. *Population Ecology of Terrestrial Orchids*.
- Wells T, Rothery P, Cox R, Bamford S. 1998. Flowering dynamics of *Orchis morio* L. and *Herminium monorchis* (L.) R. Br. at two sites in eastern England. *Botanical Journal of the Linnean Society* 126:39-48.

- Wells TC, Willems JH. 1991. Population ecology of terrestrial orchids. In: *International Orchid Symposium (1990: South Limburg, Netherlands)*. SPB Academic Publishing,. DOI: [90-5103-068-1](https://doi.org/10.1007/978-1-4020-0681-1).
- Werner PA, Caswell H. 1977. Population growth rates and age versus stage-distribution models for teasel (*Dipsacus sylvestris* Huds.). *Ecology* 58:1103-1111.
- Werner PA, Peacock SJ. 2019. Savanna canopy trees under fire: long-term persistence and transient dynamics from a stage-based matrix population model. *Ecosphere* 10:e02706.
- Whigham DF, O'Neill JP, Rasmussen HN, Caldwell BA, McCormick MK. 2006. Seed longevity in terrestrial orchids—potential for persistent in situ seed banks. *Biological Conservation* 129:24-30.
- Wilcock C, Neiland R. 2002. Pollination failure in plants: why it happens and when it matters. *Trends in Plant Science* 7:270-277.
- Willems J, Bik L. 1991. Population biology of *Orchis simia* in the Netherlands, 1972–1990. In: Wells TCE, Willems JH eds. *Population Ecology of Terrestrial Orchids*. SPB Academic Publishing bv, The Hague, 33-45.
- Willems J, Dorland E. 2000. Flowering frequency and plant performance and their relation to age in the perennial orchid *Spiranthes spiralis* (L.) Chevall. *Plant Biology* 2:344-349.
- Williams JL, Ellis MM, Bricker MC, Brodie JF, Parsons EW. 2011. Distance to stable stage distribution in plant populations and implications for near-term population projections. *Journal of Ecology* 99:1171-1178.
- Willmer P. 2011. Pollination and floral ecology. In: *Pollination and floral ecology*. Princeton University Press,.
- Winkler M, Hietz P. 2001. Population structure of three epiphytic orchids (*Lycaste aromatica*, *Jacquiniella leucomelana*, and *J. teretifolia*) in a Mexican humid montane forest. *Selbyana*:27-33.
- Winkler M, Hülber K, Hietz P. 2009. Population dynamics of epiphytic orchids in a metapopulation context. *Annals of Botany* 104:995-1004. DOI: [10.1093/aob/mcp188](https://doi.org/10.1093/aob/mcp188).
- Wright SJ, Muller-Landau HC, Condit R, Hubbell SP. 2003. Gap-dependent recruitment, realized vital rates, and size distributions of tropical trees. *Ecology* 84:3174-3185.
- Young K, Ewel J, Brown B. 1987. Seed dynamics during forest succession in Costa Rica. *Vegetatio* 71:157-173.
- Zettler LW, McInnis J TM. 1993. Symbiotic seed germination and development of *Spiranthes cernua* and *Goodyera pubescens* (Orchidaceae: Spiranthoideae).
- Zhang X, Jia Y, Liu Y, Chen D, Luo Y, Niu S. 2021. Challenges and perspectives in the study of self-incompatibility in orchids. *International Journal of Molecular Sciences* 22:12901.
- Zhongjian L, Lijun C, Wenhui R, Liqiang L, Yuting Z. 2008. Correlation between numerical dynamics and reproductive behavior in *Cypripedium lentiginosum*. *Acta Ecologica Sinica* 28:111-121.
- Zotz G. 1995. How fast does an epiphyte grow? *Selbyana*:150-154.
- Zotz G. 1998. Demography of the epiphytic orchid, *Dimerandra emarginata*. *Journal of Tropical Ecology* 14:725-741.
- Zotz G, Schmidt G. 2006. Population decline in the epiphytic orchid *Aspasia principissa*. *Biological Conservation* 129:82-90. DOI: [10.1016/j.biocon.2005.07.022](https://doi.org/10.1016/j.biocon.2005.07.022).